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4.1 International Economics and Trade

1. Globalisation

Globalisation is the increasing integration and interdependence of the world's economies through the growth of international trade, investment, migration, and the spread of technology and culture across borders.

Causes of Globalisation

- **Reduced trade barriers** — tariff reductions through GATT/WTO rounds (average tariffs fell from ~40% in 1947 to ~3% today).
- **Improved transport** — containerisation cut shipping costs by over 90%; air freight enables just-in-time global supply chains.
- **Advances in ICT** — internet, fibre optics, and mobile technology enable instant communication and e-commerce across borders.
- **Deregulation of financial markets** — removal of capital controls allows free movement of money between countries.
- **Growth of trading blocs** — EU, NAFTA/USMCA, ASEAN reduce barriers between member states.
- **Growth of transnational corporations (TNCs)** — firms like Apple, Toyota, and Unilever operate supply chains spanning dozens of countries.
- **Political change** — collapse of the Soviet Union and China's opening up (post-1978 reforms) brought billions into the global market economy.

Consequences of Globalisation

Benefits	Costs

Greater consumer choice and lower prices from competition	Structural unemployment in industries exposed to cheaper imports (e.g. UK steel, US manufacturing)
Economies of scale for firms accessing larger markets	Widening inequality — gains concentrated among skilled workers and capital owners
Technology transfer to developing countries raises productivity	Environmental degradation — increased carbon emissions from transport; race to the bottom on regulation
Reduced absolute poverty — over 1 billion lifted out of extreme poverty since 1990	Cultural homogenisation — loss of local traditions and languages
FDI brings capital, jobs, and management expertise to host nations	Tax avoidance by TNCs shifting profits to low-tax jurisdictions
Increased specialisation based on comparative advantage	Contagion risk — financial crises spread quickly (e.g. 2008 Global Financial Crisis)

Evaluation: Globalisation is not inherently good or bad. The key issue is the *distribution* of gains. Developing countries with good governance and institutions (e.g. South Korea, Vietnam) have benefited greatly, while those with weak institutions or primary product dependency (e.g. some Sub-Saharan African nations) have seen fewer gains. Government policy — such as retraining programmes, progressive taxation, and environmental regulation — is crucial to ensuring the benefits are widely shared.

2. Absolute and Comparative Advantage

Absolute Advantage (Adam Smith, 1776)

Absolute advantage: A country has an absolute advantage in producing a good if it can produce it using fewer resources (or produce more per unit of input) than another country.

If Country A is better at producing wine and Country B is better at producing cloth, each should specialise in its strength and trade. Both consume more than under autarky (self-sufficiency).

Comparative Advantage (David Ricardo, 1817)

> **Comparative advantage:** A country has a comparative advantage in a good if it can produce it at a *lower opportunity cost* than another country. Even if one country has an absolute advantage in both goods, mutually beneficial trade is still possible.

Numerical Example

Suppose two countries can produce the following per worker per day:

Country	Wine (bottles)	Cloth (metres)
France	6	3
England	2	2

Opportunity costs:

- **France:** 1 bottle of wine costs 3/6 = **0.5 metres** of cloth. 1 metre of cloth costs 6/3 = **2 bottles** of wine.
- **England:** 1 bottle of wine costs 2/2 = **1 metre** of cloth. 1 metre of cloth costs 2/2 = **1 bottle** of wine.

France has the lower opportunity cost of wine (0.5 < 1), so France has a **comparative advantage in wine**.

England has the lower opportunity cost of cloth (1 < 2), so England has a **comparative advantage in cloth**.

If France specialises in wine and England in cloth, total world output increases and both can consume beyond their PPFs through trade.

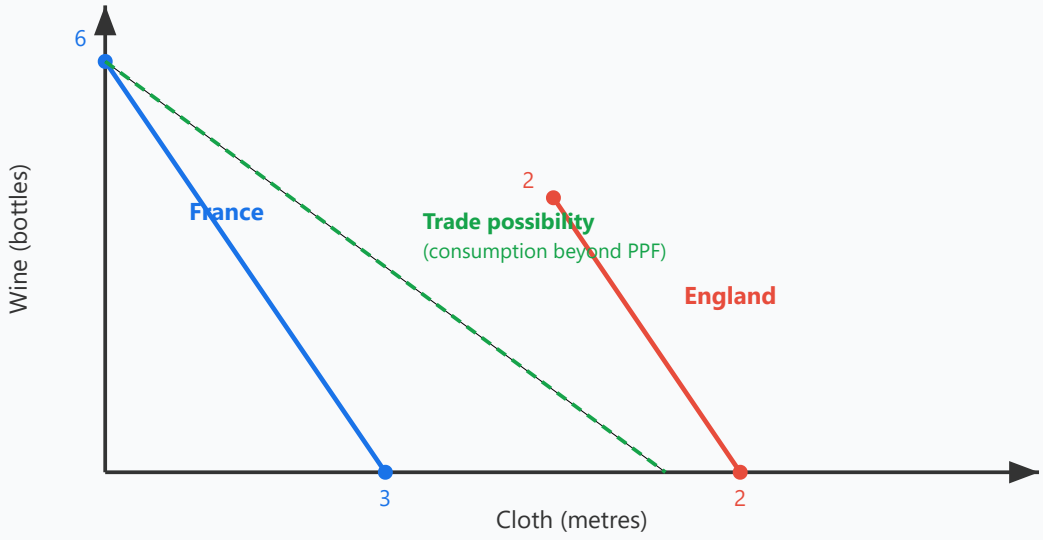

Figure 1: PPFs for France and England showing gains from trade

Assumptions and Limitations of Comparative Advantage

- **Constant costs of production** — in reality, opportunity costs tend to increase as resources are reallocated (concave PPF).
- **No transport costs** — in practice, these can erode the gains from trade.
- **Perfect factor mobility within countries** — workers displaced from declining industries may struggle to retrain.
- **No externalities** — trade can cause environmental damage not captured in the model.
- **Two countries, two goods** — the real world is far more complex.
- **Free trade assumed** — tariffs, quotas, and non-tariff barriers distort outcomes.
- **Static model** — does not account for dynamic comparative advantage where countries develop new strengths over time (e.g. South Korea moving from textiles to semiconductors).

Exam Tip: Always state the assumptions you are making when using a comparative advantage model. Evaluation marks come from questioning whether these assumptions hold in the real world.

3. Terms of Trade

Terms of trade: The ratio of a country's average export prices to its average import prices, expressed as an index number.

$$\text{Terms of Trade} = (\text{Index of Export Prices} / \text{Index of Import Prices}) \text{ x } 100$$

- A **rise** in the terms of trade (improvement) means export prices rise relative to import prices. Fewer exports are needed to buy the same volume of imports.
- A **fall** in the terms of trade (deterioration) means export prices fall relative to import prices. More exports are needed to buy the same volume of imports.

Factors Affecting Terms of Trade

- **Exchange rate** — an appreciation raises export prices in foreign currency (improves ToT) but may reduce export competitiveness.
- **Relative inflation rates** — higher domestic inflation raises export prices, improving ToT but worsening competitiveness.
- **Changes in world commodity prices** — rising oil prices improve ToT for oil exporters (e.g. Saudi Arabia) but worsen it for importers (e.g. India).
- **Productivity changes** — rising productivity can lower export prices (worsening ToT) but increase volume and competitiveness.
- **Changes in demand for exports/imports** — growing demand for a country's exports improves its ToT.

Evaluation: An improvement in the terms of trade is not always beneficial. If it results from rising export prices due to cost-push inflation, export volumes may fall, worsening the current account. The impact depends on the price elasticity of demand for exports and imports (see Marshall-Lerner condition).

4. Free Trade: Benefits and Costs

Benefits of Free Trade

- **Allocative efficiency** — resources flow to where they are most productively used, guided by comparative advantage.
- **Lower prices and greater choice** for consumers due to increased competition.
- **Economies of scale** — access to larger markets allows firms to reduce average costs.
- **Dynamic efficiency** — competitive pressure incentivises innovation and R&D.
- **Technology transfer** — FDI and trade spread knowledge and best practices.
- **Higher economic growth** — empirical evidence shows strong correlation between trade openness and growth rates.

Costs of Free Trade

- **Structural unemployment** — workers in uncompetitive industries lose jobs (e.g. US manufacturing decline).
- **Infant industry vulnerability** — new domestic industries cannot compete against established foreign producers.
- **Over-specialisation risk** — dependence on a narrow range of exports leaves countries vulnerable to demand/price shocks.
- **Environmental concerns** — "race to the bottom" as countries lower environmental standards to attract investment.
- **Worsening inequality** — unskilled workers in developed countries may see falling real wages.
- **Loss of sovereignty** — countries may lose ability to set independent industrial/environmental policy.
- **Dumping** — foreign firms sell below cost to destroy domestic competition, then raise prices.

5. Trade Barriers (Protectionism)

Protectionism: Government policies that restrict international trade to shield domestic industries from foreign competition.

Arguments for Protectionism

- **Infant industry argument** — new industries need temporary protection until they achieve economies of scale and can compete globally.

- **Protect strategic industries** — defence, energy, and food production may need domestic capacity for national security.
- **Prevent dumping** — anti-dumping tariffs counteract foreign firms selling below cost.
- **Protect employment** — prevents sudden job losses in vulnerable sectors.
- **Correct a current account deficit** — restricting imports can improve the trade balance.
- **Raise government revenue** — particularly important for developing countries with limited tax bases.
- **Protect health and safety standards** — prevent imports of goods that do not meet domestic regulations.
- **Prevent "race to the bottom"** — maintain environmental and labour standards.

Types of Trade Barriers

a) Tariffs

Tariff: A tax imposed on imported goods, raising their price in the domestic market.

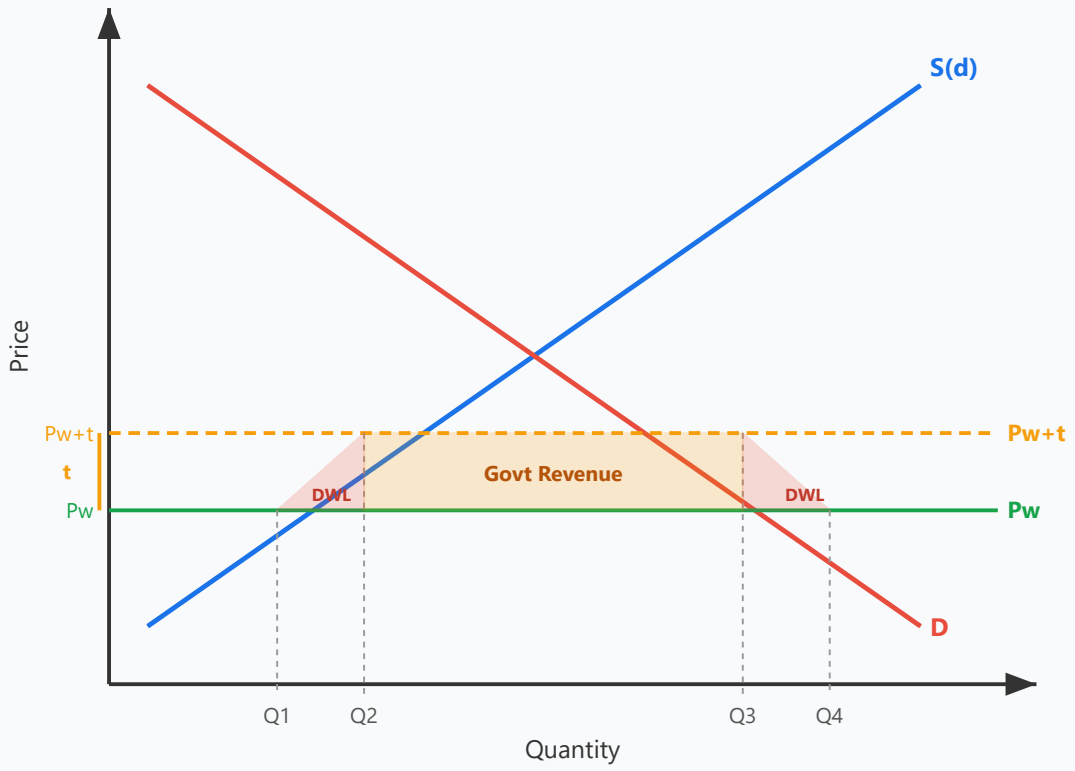

Figure 2: Effect of a tariff on an imported good

Effects of a tariff:

- **Price rises** from Pw to Pw+t.
- **Domestic output rises** from Q1 to Q2 (domestic producers can now compete at the higher price).
- **Imports fall** from (Q4-Q1) to (Q3-Q2).
- **Consumer surplus falls** — consumers pay higher prices and buy less.
- **Producer surplus rises** — domestic firms receive higher prices.
- **Government revenue** = tariff x quantity of imports = t x (Q3-Q2).
- **Deadweight welfare loss** — two triangles representing production inefficiency (resources used in higher-cost domestic production) and consumption inefficiency (consumers priced out of the market).

b) Quotas

> **Quota:** A physical limit on the quantity of a good that can be imported in a given time period.

- Raises the domestic price (similar to a tariff).
- Creates **no government revenue** (unlike a tariff) — the economic rent goes to whoever holds the import licences.
- Creates deadweight welfare loss and reduces consumer surplus.
- More distortionary than tariffs because they create no revenue and are harder to negotiate away.

c) Embargoes

- A **complete ban** on trade with a particular country or in a particular product.
- Usually politically motivated (e.g. US embargo on Cuba since 1962; EU sanctions on Russia post-2022).
- Eliminates all trade in the affected goods — maximum protectionist measure.

d) Non-Tariff Barriers

- **Administrative barriers** — complex customs procedures, licensing requirements, excessive paperwork.
- **Product standards / regulations** — setting technical standards that foreign producers find difficult to meet.
- **Subsidies to domestic producers** — enable domestic firms to undercut foreign competitors (e.g. EU Common Agricultural Policy subsidies to European farmers).

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- **Government procurement policies** — favouring domestic firms in public contracts.
- **Exchange rate manipulation** — keeping currency artificially low to make exports cheaper (China was frequently accused of this).

Evaluation of protectionism: While infant industry protection has historical precedent (South Korea, Japan), it risks creating permanent dependence on protection if industries never become competitive. Protectionism often invites retaliation, reducing trade for all. The WTO estimates that a full-scale trade war could reduce global GDP by up to 5%. Consumer welfare almost always falls in the short run.

Case Study: US-China Trade War (2018-present)

The US imposed tariffs on over $360 billion of Chinese imports (up to 25%) citing unfair trade practices, IP theft, and the trade deficit. China retaliated with tariffs on ~$110 billion of US goods (soybeans, cars, aircraft). Effects included: higher prices for US consumers (estimated $800-1300 per household per year), disrupted global supply chains, reduced US farm exports requiring $28 billion in government bailouts, accelerated "decoupling" of US-China tech sectors, and some reshoring of manufacturing. The trade deficit with China actually widened initially. This illustrates the limitations of using tariffs to correct bilateral trade imbalances and the risk of escalation.

6. Trading Blocs

Trading bloc: A group of countries that agree to reduce or eliminate trade barriers between themselves.

Types of Trading Blocs (in increasing order of integration)

Type	Features	Example
Free Trade Area	No tariffs/quotas between members. Each member keeps its own external tariffs against non-members.	USMCA (US-Mexico-Canada), EFTA

Customs Union	Free trade between members PLUS a common external tariff (CET) against non-members.	EU Customs Union, MERCOSUR
Common Market	Customs union PLUS free movement of labour and capital between members.	EU Single Market (pre-full union), EEA
Economic Union	Common market PLUS harmonised economic policies (fiscal, monetary, regulatory).	EU (with single currency eurozone), partial fiscal coordination

Trade Creation vs Trade Diversion

Trade creation: When a trading bloc leads to a shift from a higher-cost domestic producer to a lower-cost member producer. This *increases* welfare.

Trade diversion: When a trading bloc leads to a shift from a lower-cost non-member producer to a higher-cost member producer (because the CET makes the non-member's goods more expensive). This *reduces* welfare.

A trading bloc is welfare-enhancing overall if trade creation exceeds trade diversion.

Case Study: The European Union

The EU is the world's largest single market with 27 member states and ~450 million consumers. Benefits include: elimination of tariffs between members, free movement of workers (estimated 3.8% of EU working-age population lives in a different member state), harmonised product standards reducing trade costs, and the euro eliminating exchange rate risk for 20 eurozone members. Costs include: loss of independent monetary policy for eurozone members (problematic during the 2010-12 sovereign debt crisis when Greece needed devaluation but could not), net budget contributions for wealthier members, regulatory burden on businesses, and political tensions over sovereignty. Studies estimate the EU single market has increased intra-EU trade by 50-70%.

Case Study: Brexit Trade Effects

The UK left the EU single market and customs union on 31 January 2020 (transition ended December 2020). Effects include: new customs checks adding an estimated 6-8% cost burden on

UK-EU trade, UK-EU goods trade falling ~15% below the counterfactual trend by 2022, new non-tariff barriers (rules of origin, regulatory divergence, phytosanitary checks), disruption to financial services (London lost some euro-denominated clearing), reduced EU worker migration to the UK (contributing to labour shortages in hospitality, agriculture, and logistics), but the UK gained freedom to sign independent trade deals (e.g. UK-Australia FTA, UK-Japan CEPA). The OBR estimated Brexit would reduce UK GDP by ~4% in the long run compared to remaining in the EU.

7. The World Trade Organisation (WTO)

WTO: An international organisation (established 1995, successor to GATT) with 164 member countries, responsible for setting rules for international trade and resolving trade disputes.

Key Functions

- **Negotiate trade liberalisation** — multilateral trade rounds aim to reduce tariffs and non-tariff barriers (e.g. Uruguay Round, Doha Round).
- **Dispute settlement** — a binding legal process to resolve trade conflicts between members (e.g. Boeing-Airbus subsidies dispute).
- **Monitor trade policies** — regular reviews of members' trade policies promote transparency.
- **Most Favoured Nation (MFN) principle** — members must treat all other WTO members equally (cannot give one country a better tariff rate without giving it to all, except within trading blocs).
- **Technical assistance** to developing countries to help them participate in global trade.

Limitations of the WTO

- **Doha Round stalemate** — launched 2001, still incomplete. Developing and developed countries cannot agree on agriculture subsidies and market access.
- **Slow decision-making** — consensus required among 164 members.
- **Enforcement challenges** — dispute resolution can take years; powerful countries may ignore rulings.
- **Rise of bilateral/regional deals** — countries increasingly bypass the WTO to form their own agreements.
- **Criticism from developing countries** — rules seen as favouring developed nations (e.g. agricultural subsidies in the EU and US distort markets).

Case Study: China's WTO Accession (2001)

China joined the WTO in December 2001 after 15 years of negotiations. Effects: China's exports grew from $266 billion (2001) to over $3.5 trillion (2022), becoming the world's largest goods exporter. China cut average tariffs from 43% to ~7.5%. Benefits included massive poverty reduction (~800 million lifted out of extreme poverty) and lower consumer prices worldwide. Costs included: the "China shock" — an estimated 2-2.4 million US manufacturing jobs lost (2000-2015), allegations of currency manipulation and state subsidies violating WTO rules, and concerns about intellectual property protection. China's experience illustrates both the enormous gains from trade integration and the distributional challenges that accompany it.

8. Exchange Rates

Exchange rate: The price of one currency expressed in terms of another (e.g. GBP1 = USD1.27).

Exchange Rate Systems

System	How It Works	Advantages	Disadvantages
Floating	Determined entirely by market forces of supply and demand. No government intervention. (e.g. GBP, USD, EUR)	Automatic adjustment to external shocks; monetary policy independence; no need for large forex reserves	Volatility creates uncertainty for trade and investment; can overshoot; speculative attacks
Fixed	Government/central bank pegs the currency at a set rate against another currency. Uses forex reserves to buy/sell currency to maintain the peg. (e.g. Hong Kong dollar pegged to USD)	Certainty for trade and investment; disciplines monetary policy; reduces speculation	Requires large forex reserves; loss of monetary policy independence; may become misaligned with fundamentals

Managed float	Mainly market-determined but central bank intervenes occasionally to smooth excessive volatility or prevent misalignment. (e.g. India, China)	Combines flexibility with some stability; retains some policy independence	Uncertainty about when/how intervention occurs; can be seen as manipulation; still requires reserves

Determination of Floating Exchange Rates

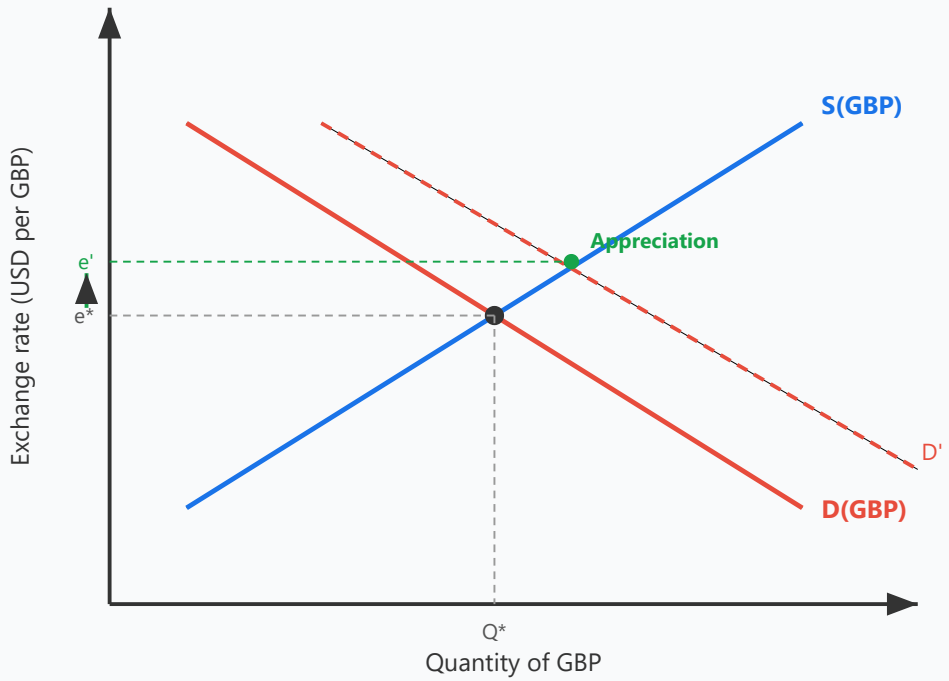

Figure 3: An increase in demand for GBP causes an appreciation (e to e')*

Factors Affecting Exchange Rates

Demand for a currency increases (appreciation) when:

- Exports rise (foreigners need domestic currency to buy exports)
- Foreign direct investment inflows increase
- Interest rates rise (hot money inflows from international investors seeking higher returns)
- Speculation — traders expect the currency to appreciate, so they buy now
- Relative inflation is lower — goods remain competitive, maintaining export demand
- Economic growth prospects improve — attracting investment

Supply of a currency increases (depreciation) when:

- Imports rise (domestic consumers sell domestic currency to buy foreign goods)
- Capital outflows increase (investment abroad)
- Interest rates fall relative to other countries
- Inflation rises relative to trading partners
- Loss of confidence / political instability (e.g. GBP fell ~10% after the 2016 Brexit referendum)
- Central bank sells its own currency (intervention to weaken)

Effects of Appreciation

- Exports become **more expensive** in foreign currency — export volumes tend to fall.
- Imports become **cheaper** in domestic currency — import volumes tend to rise.
- **Current account worsens** (assuming Marshall-Lerner condition holds).
- **Lower imported inflation** — cheaper raw materials and finished goods.
- **Lower AD** (net exports fall) — may reduce growth and employment.
- Firms face competitive pressure — may improve productivity or relocate production abroad.

Effects of Depreciation

- Exports become **cheaper** in foreign currency — export volumes tend to rise.
- Imports become **more expensive** — import volumes tend to fall.
- **Current account improves** (if Marshall-Lerner holds, and after J-curve adjustment).
- **Higher imported inflation** — costlier raw materials feed into CPI.
- **Higher AD** (net exports rise) — can boost growth but also inflationary pressure.
- Helps tourism sector — the country becomes a cheaper destination.

Memory Aid — SPICED:
Strong **P**ound: **I**mports **C**heaper, **E**xports **D**earer
(And the reverse: Weak Pound — Imports Dearer, Exports Cheaper — WIDEC)

9. Marshall-Lerner Condition

Marshall-Lerner condition: A depreciation of the exchange rate will improve the current account (reduce a deficit) only if the sum of the price elasticity of demand for exports (PED_X) and the price elasticity of demand for imports (PED_M) is greater than 1.

$$|\mathrm{PED_X}| + |\mathrm{PED_M}| > 1 \rightarrow \text{Depreciation improves the current account}$$

Explanation: When a currency depreciates, export prices fall and import prices rise in domestic currency terms. Whether the trade balance improves depends on how responsive demand is to these price changes:

- If demand for exports and imports is **price elastic** (M-L holds), the volume changes are large enough to outweigh the adverse price effects, so the trade balance improves.
- If demand is **price inelastic** (M-L does not hold), volumes change little, but import prices are higher, so the trade balance *worsens*.

Evaluation: In the short run, demand for imports and exports tends to be price inelastic (contracts are fixed, consumers take time to adjust), so the M-L condition often does not hold initially. Over time, elasticities increase as consumers and firms adjust — this gives rise to the J-curve effect.

10. The J-Curve Effect

J-curve effect: Following a depreciation, the current account initially worsens before eventually improving, tracing out a path shaped like the letter J.

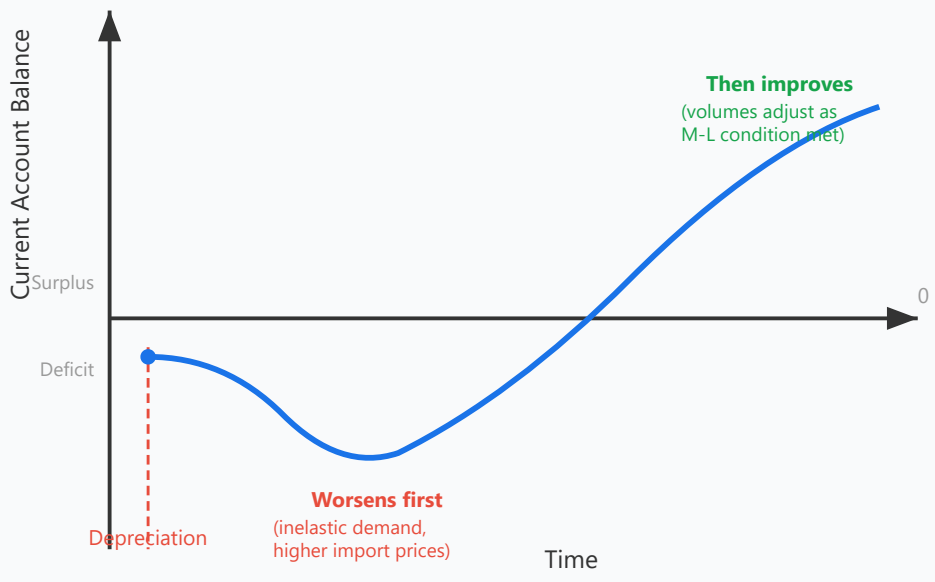

Figure 4: The J-Curve — current account worsens before improving after depreciation

Why does the current account worsen initially?

- In the **short run**, import and export volumes are slow to change (existing contracts, consumer habits, time to find new suppliers).
- Import prices immediately rise in domestic currency terms, so the import bill increases even though volumes have not yet fallen.
- Over the **medium to long run**, consumers and firms respond to the new relative prices: export volumes rise, import volumes fall, and the current account improves.

Exam Tip: The J-curve is a key evaluation point when discussing the effectiveness of depreciation as a policy to correct a current account deficit. Always note the time lag.

11. Balance of Payments

Balance of payments (BoP): A record of all financial transactions between one country and the rest of the world over a given period. It must always balance (current account + capital account + financial account + errors and omissions = 0).

Components

Account	What It Records	Examples
Current Account	Trade in goods, trade in services, primary income (investment income, wages), secondary income (transfers)	UK exports of financial services, oil imports, dividends from overseas investment, foreign aid payments
Capital Account	Capital transfers, purchase/sale of non-produced assets	Debt forgiveness, purchase of patents, EU structural funds (small)
Financial Account	FDI, portfolio investment, reserve assets, other investment	Toyota building a factory in the UK (FDI inflow), UK pension fund buying US bonds (portfolio outflow), Bank of England forex reserves

Current Account Deficit: Causes

- **Overvalued exchange rate** — makes exports expensive and imports cheap.
- **High relative inflation** — domestic goods become uncompetitive.
- **Strong domestic growth** — rising incomes suck in imports (high income elasticity of demand for imported goods).
- **Low productivity** relative to competitors — higher unit labour costs.
- **Structural factors** — deindustrialisation, loss of manufacturing base, over-reliance on imported energy or consumer goods.
- **Non-price factors** — poor quality, design, or after-sales service of domestic goods.

Significance of a Current Account Deficit

A current account deficit may not be a problem if:

- It reflects strong economic growth (high incomes pulling in imports) — a sign of prosperity.
- It is financed by sustainable capital inflows (FDI) rather than short-term hot money or borrowing.
- It is small relative to GDP (e.g. 1-2% of GDP is generally manageable).
- Investment inflows are going into productive capacity that will generate future export earnings.

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A current account deficit IS a problem if:

- It is large and persistent — signals deep-rooted competitiveness problems.
- It is financed by short-term borrowing or hot money — risk of sudden capital flight and currency crisis.
- It leads to rising external debt — debt servicing costs worsen the primary income balance.
- It causes depreciation that feeds into inflation.
- It reflects structural decline in key export sectors.

Policies to Correct a Current Account Deficit

Expenditure-reducing policies:

- Deflationary fiscal/monetary policy — reduces AD, lowers incomes, reduces import demand. But causes unemployment and lower growth.

Expenditure-switching policies:

- Depreciation/devaluation — makes exports cheaper and imports dearer (subject to Marshall-Lerner and J-curve).
- Tariffs and quotas — directly reduce imports but risk retaliation and WTO violations.

Supply-side policies:

- Investment in education, training, R&D, infrastructure — improves long-run competitiveness. Most effective but slowest to take effect.

12. International Competitiveness

International competitiveness: The ability of a country's firms to compete successfully in international markets, reflected in price and non-price factors.

Measures of Competitiveness

- **Relative unit labour costs (ULCs)** — wages adjusted for productivity. Lower ULCs = more competitive.
- **Relative export prices** — the price of a country's exports compared to competitors' exports.

- **Real effective exchange rate (REER)** — trade-weighted exchange rate adjusted for relative inflation.
- **Current account position** — persistent surpluses may indicate strong competitiveness.
- **Global competitiveness indices** — e.g. World Economic Forum rankings considering institutions, infrastructure, innovation, etc.

Factors Affecting Competitiveness

Price factors:

- Exchange rate level
- Relative inflation rates
- Labour costs and productivity (unit labour costs)
- Raw material / energy costs
- Tax burden on businesses

Non-price factors:

- Quality and reliability of products
- Innovation and technology (R&D spending, patents)
- Design, branding, and marketing
- After-sales service and customer support
- Infrastructure quality (transport, digital)
- Education and skills of the workforce (human capital)
- Institutional quality — rule of law, property rights, ease of doing business
- Flexibility of labour markets

Evaluation: Price competitiveness can be quickly affected by exchange rate movements, but non-price competitiveness is more sustainable in the long run. Germany maintains a large trade surplus despite relatively high wages because of strong non-price competitiveness (engineering quality, brand reputation, innovation). Supply-side policies to improve non-price competitiveness take years to bear fruit but are the most effective long-term strategy.

Exam Tip: When evaluating any policy to improve competitiveness or the current account, always consider: (1) time lags, (2) the Marshall-Lerner condition, (3) whether the problem is cyclical or

structural, (4) possible side effects (inflation, unemployment), (5) risk of retaliation if protectionist measures are used.